

MODUL 12 & MODUL 13

Laravel : Database 1 & Database 2

Modul 12

- a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

➤ env

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=ecommerce
DB_USERNAME=root
DB_PASSWORD=
DB_CHARSET=utf8mb4
DB_COLLATION=utf8mb4_unicode_ci
```

➤ Migration products_table

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('products', function (Blueprint $table) {
            $table->id();
            $table->timestamps();
        });
    }
}
```

```

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('products');
    }
};

```

➤ Product.php

```

<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    use HasFactory;

    // Menentukan kolom yang boleh diisi secara massal
    protected $fillable = ['name', 'price'];
}

```

➤ ProductController.php

```

<?php

namespace App\Http\Controllers;

use App\Models\Product;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    // Menampilkan daftar produk
    public function index()
    {
        $products = Product::all();
        return view('products.index', compact('products'));
    }
}

```

```
// Menampilkan form tambah produk
public function create()
{
    return view('products.form', ['title' => 'Tambah']);
}

// Menyimpan data produk baru dengan validasi
public function store(Request $request)
{
    $request->validate([
        'name' => 'required|min:4',
        'price' => 'required|integer|min:1000000',
    ]);

    Product::create($request->all()); // Mass assignment
    return redirect()->route('products.index')->with('success',
'Produk berhasil ditambahkan!');
}

// Menampilkan form edit produk
public function edit(Product $product)
{
    return view('products.form', ['title' => 'Edit', 'product' =>
$product]);
}

// Memperbarui data produk dengan validasi
public function update(Request $request, Product $product)
{
    $request->validate([
        'name' => 'required|min:4',
        'price' => 'required|integer|min:1000000',
    ]);

    $product->update($request->all());

    return redirect()->route('products.index')->with('success',
'Produk berhasil diperbarui!');
}

// Menghapus data produk
public function destroy(Product $product)
{
    $product->delete();
    return redirect()->route('products.index')->with('success',
```

```
'Produk berhasil dihapus!');  
    }  
}
```

➤ web.php

```
<?php  
  
use Illuminate\Support\Facades\Route;  
use App\Http\Controllers\ProductController;  
  
// Route CRUD Resource  
Route::resource('products', ProductController::class);  
  
// Contoh Route Session handling (Logout)  
Route::get('/logout', function () {  
    session()->flush(); // Menghapus seluruh session  
    return redirect('/login');  
});
```

➤ template.blade.php

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
    <meta charset="UTF-8">  
    <meta name="viewport" content="width=device-width, initial-  
scale=1.0">  
    <title>@yield('title') - Aplikasi Web</title>  
    <link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstr  
ap.min.css" rel="stylesheet">  
</head>  
<body class="bg-light">  
    <div class="container py-5">  
        @yield('content')  
    </div>  
</body>  
</html>
```

➤ index.blade.php

```
@extends('layouts.template')  
  
@section('title', 'Daftar Produk')
```

```

@section('content')
<div class="row justify-content-center">
    <div class="col-md-10">
        @if (session('success'))
            <div class="alert alert-
success">{{ session('success') }}</div>
        @endif

        <div class="d-flex justify-content-between align-items-
center mb-4">
            <h2>Daftar Produk</h2>
            <a href="{{ route('products.create') }}" class="btn
btn-primary">Tambah Produk</a>
        </div>

        <table class="table table-bordered table-striped bg-white
shadow-sm">
            <thead>
                <tr>
                    <th>Nama Produk</th>
                    <th>Harga</th>
                    <th>Aksi</th>
                </tr>
            </thead>
            <tbody>
                @foreach ($products as $product)
                    <tr>
                        <td>{{ $product->name }}</td>
                        <td>Rp {{ number_format($product->price, 0,
',', '.') }}</td>
                        <td>
                            <a href="{{ route('products.edit',
$product->id) }}" class="btn btn-warning btn-sm">Edit</a>
                            <form
action="{{ route('products.destroy', $product->id) }}"
method="POST" class="d-inline">
                                @csrf
                                @method('DELETE')
                                <button type="submit" class="btn
btn-danger btn-sm" onclick="return confirm('Yakin ingin menghapus
data ini?')">Hapus</button>
                            </form>
                        </td>
                    </tr>
                @endforeach
            </tbody>
        </table>
    </div>
</div>

```

```

        @endforeach
    </tbody>
</table>
</div>
</div>
@endsection

```

➤ form.blade.php

[illegible]

```

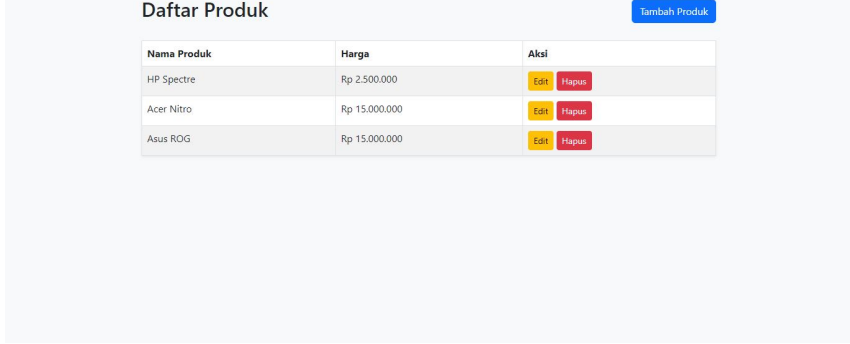
class="form-control @error('price') is-invalid @enderror"
id="price" value="{{ old('price', $product->price ?? '') }}"
        @error('price')
            <div class="invalid-
feedback">{{ $message }}</div>
        @enderror
    </div>

    <div class="d-flex justify-content-between">
        <a href="{{ route('products.index') }}"
class="btn btn-secondary">Kembali</a>
        <button type="submit" class="btn btn-
success">Simpan Data</button>
    </div>
</form>
</div>
</div>
</div>
</div>
</div>
@endsection

```

b. Screenshot output

<http://localhost:8000/products>



<http://localhost:8000/products/create>

Tambah Produk

Nama Produk

Laptop

Harga (Minimum Rp1.000.000)

10000000

Kembali

Simpan Data

<http://localhost:8000/products>

Produk berhasil ditambahkan!

Daftar Produk

Tambah Produk

Nama Produk	Harga	Aksi	
HP Spectre	Rp 2.500.000	Edit	Hapus
Acer Nitro	Rp 15.000.000	Edit	Hapus
Asus ROG	Rp 15.000.000	Edit	Hapus
Laptop	Rp 10.000.000	Edit	Hapus

<http://localhost:8000/9/edit>

Edit Produk

Nama Produk

Laptop

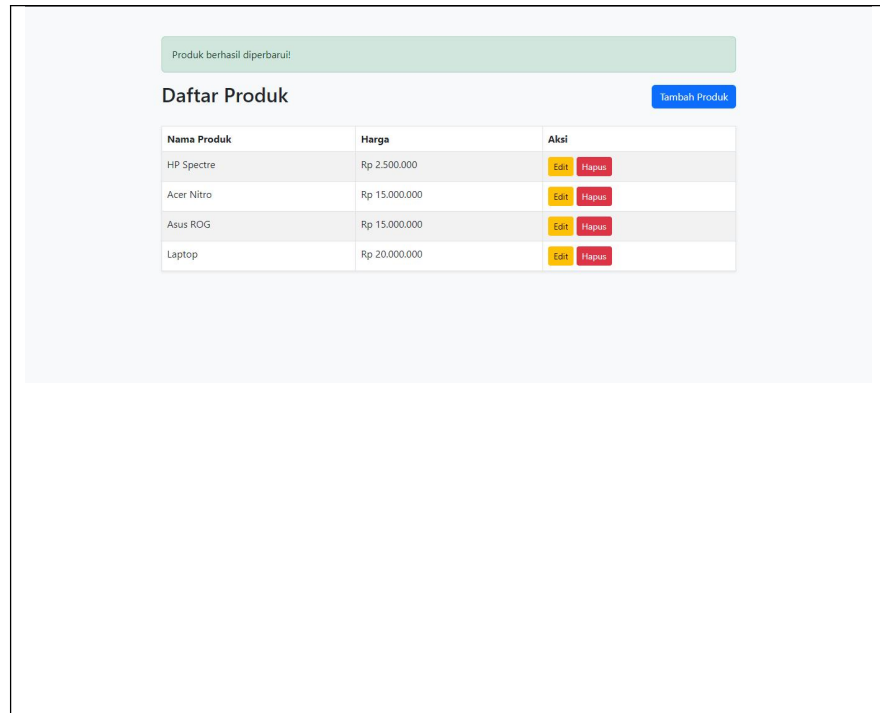
Harga (Minimum Rp1.000.000)

20000000

Kembali

Simpan Data

<http://localhost:8000/products>



- c. Penjelasan dari code dan output (korelasi)
- Arsitektur *Model-View-Controller* (MVC) berkolaborasi untuk mengelola operasi CRUD dasar pada tabel products di dalam database ecommerce. Fungsi `index()` pada *controller* menginstruksikan Eloquent Model untuk menarik semua baris data produk, lalu mengirimkannya ke file tampilan (`index.blade.php`) untuk dirender secara dinamis menggunakan tabel HTML berdesain Bootstrap di browser pengguna. Ketika melakukan manipulasi data, fungsi `store()` dan `update()` pada *controller* akan mengeksekusi aturan validasi yang ketat (seperti nama minimal 4 karakter dan harga minimal satu juta rupiah); jika aturan ini dilanggar, sistem backend akan memicu direktif `@error` di sisi view untuk menampilkan pesan kesalahan berwarna merah tepat di bawah input form. Sebaliknya, apabila data yang dimasukkan lolos validasi dan berhasil disimpan, fungsi *controller* akan mengirimkan *flash session* sukses yang langsung ditangkap oleh halaman utama untuk memunculkan kotak notifikasi alert berwarna hijau di layar.

Modul 13

a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

➤ Migration categories_table

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('categories', function (Blueprint $table) {
            $table->id();
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('categories');
    }
};
```

➤ Migration add_category_id_to_products_table

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
```

```

public function up(): void
{
    Schema::table('products', function (Blueprint $table) {
        //
    });
}
/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::table('products', function (Blueprint $table) {
        //
    });
}
};

```

➤ Migration variants_table

```

<?php
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('variants', function (Blueprint $table) {
            $table->id();
            $table->timestamps();
        });
    }
    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('variants');
    }
};

```

➤ Category.php

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Category extends Model
{
    use HasFactory;

    protected $fillable = ['category_name'];

    // Relasi One-to-Many: Kategori memiliki banyak produk
    public function products()
    {
        return $this->hasMany(Product::class);
    }
}
```

➤ Product.php

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    use HasFactory;

    protected $fillable = ['name', 'price', 'category_id']; //
    Ditambahkan category_id

    // Relasi Inverse One-to-Many: Produk dimiliki oleh satu Kategori
    public function category()
    {
        return $this->belongsTo(Category::class);
    }
}
```

➤ index.blade.php

```
@extends('layouts.template')

@section('title', 'Daftar Produk')

@section('content')
<div class="row justify-content-center">
    <div class="col-md-10">
        @if (session('success'))
            <div class="alert alert-
success">{{ session('success') }}</div>
        @endif

        <div class="d-flex justify-content-between align-items-
center mb-4">
            <h2>Daftar Produk</h2>
            <a href="{{ route('products.create') }}" class="btn
btn-primary">Tambah Produk</a>
        </div>

        <table class="table table-bordered table-striped bg-white
shadow-sm">
            <thead>
                <tr>
                    <th>Nama Produk</th>
                    <th>Harga</th>
                    <th>Kategori</th> <th>Aksi</th>
                </tr>
            </thead>
            <tbody>
                @foreach ($products as $product)
                    <tr>
                        <td>{{ $product->name }}</td>
                        <td>Rp {{ number_format($product->price, 0,
',', '.') }}</td>
                        <td>{{ $product->category->category_name ??
'Tanpa Kategori' }}</td> <td>
                            <a href="{{ route('products.edit',
$product->id) }}" class="btn btn-warning btn-sm">Edit</a>
                            <form action="{{ route('products.destroy',
$product->id) }}" method="POST" class="d-inline">
                                @csrf
                                @method('DELETE')
                                <button type="submit" class="btn btn-
```

```

danger btn-sm" onclick="return confirm('Yakin ingin menghapus data
ini?')">Hapus</button>
        </form>
    </td>
</tr>
@endforeach
</tbody>
</table>
</div>
</div>
@endsection

```

➤ SiteController.php

```

<?php

namespace App\Http\Controllers;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\Auth;
use App\Models\User;

class SiteController extends Controller
{
    // Menampilkan halaman form login
    public function login()
    {
        return view('login');
    }

    // Memproses data input login (Autentikasi)
    public function auth(Request $req)
    {
        // Menggunakan Auth::attempt sesuai standar keamanan Laravel di
        modul
        if (Auth::attempt(['email' => $req->em, 'password' => $req->
        >pwd])) {
            // Jika sukses login, arahkan ke halaman produk
            return redirect('/products');
        }

        // Jika gagal, kembalikan ke halaman login dengan pesan error
        return redirect('/login')->with('msg', 'Email atau Password
        salah!');
    }
}

```

➤ web.php

```
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\ProductController;
use App\Http\Controllers\SiteController;

// Halaman Utama otomatis ke Login atau Products
Route::get('/', function () {
    return redirect('/login');
});

// Route Autentikasi Modul 12
Route::get('/login', [SiteController::class, 'login'])->
>name('login');
Route::post('/auth', [SiteController::class, 'auth'])->
>name('auth');

// Route Logout (Menghapus session dan keluar)
Route::get('/logout', function () {
    session()->flush(); // Menghapus semua data session
    Auth::logout();      // Keluar dari sistem autentikasi
    return redirect('/login');
})->name('logout');

// Route CRUD Products
Route::resource('products', ProductController::class);
```

➤ login.blade.php

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Login - Aplikasi Web</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstr
ap.min.css" rel="stylesheet">
```

```

</head>

<body class="bg-light">
  <div class="container mt-5">
    <div class="row justify-content-center">
      <div class="col-md-4">
        <div class="card shadow-sm mt-5">
          <div class="card-header bg-primary text-white text-center">
            <h4 class="mb-0">Form Login</h4>
          </div>
          <div class="card-body">
            @if (session('msg'))
              <div class="alert alert-danger text-center">{{ session('msg') }}</div>
            @endif

            <form method="POST" action="{{ route('auth') }}">
              @csrf

              <div class="mb-3">
                <label for="em" class="form-label">Email Address</label>
                <input type="email" name="em" id="em" class="form-control" required>
              </div>

              <div class="mb-3">
                <label for="pwd" class="form-label">Password</label>
                <input type="password" name="pwd" id="pwd" class="form-control" required>
              </div>

              <button type="submit" class="btn btn-primary w-100">Sign In</button>
            </form>
          </div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>

```


➤ template.blade.php

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>@yield('title') - Aplikasi Web</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstr
ap.min.css" rel="stylesheet">
</head>

<body class="bg-light">

    <nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4">
        <div class="container">
            <a class="navbar-brand"
href="{{ route('products.index') }}">Dashboard</a>
            <button class="navbar-toggler" type="button" data-bs-
toggle="collapse" data-bs-target="#navbarNav">
                <span class="navbar-toggler-icon"></span>
            </button>
            <div class="collapse navbar-collapse" id="navbarNav">
                <ul class="navbar-nav ms-auto">
                    <li class="nav-item me-2">
                        <a class="nav-link text-white"
href="{{ route('products.index') }}">Daftar Produk</a>
                    </li>
                    <li class="nav-item">
                        <a class="nav-link btn btn-danger btn-sm text-white px-
3" href="{{ route('logout') }}"
                            onclick="return confirm('Apakah Anda yakin ingin
keluar?')">Log Out</a>
                    </li>
                </ul>
            </div>
        </div>
    </nav>

    <div class="container py-3">
        @yield('content')
```

```

</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap
.bundle.min.js"></script>
</body>

</html>

```

b. Screenshot output

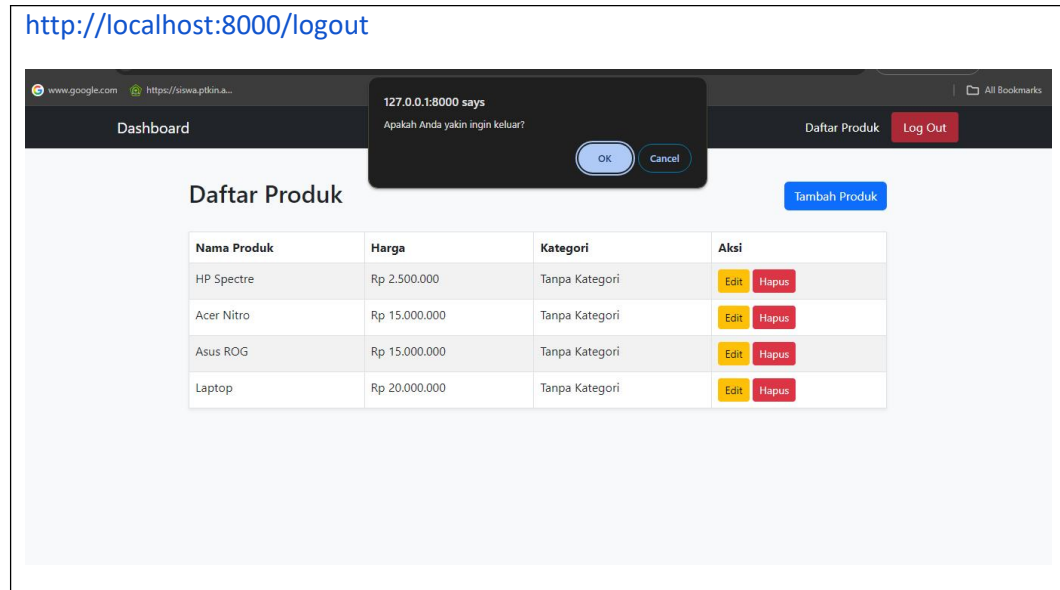
<http://localhost:8000/login>

The screenshot shows a web page with a light blue background. In the center, there is a white box with a blue header titled "Form Login". Inside the box, there are two input fields: "Email Address" with the value "Fariz@gmail.com" and "Password" with masked characters "*****". Below the password field is a blue button labeled "Sign In".

<http://localhost:8000/products>

The screenshot shows a dashboard with a dark blue header. On the left, it says "Dashboard". On the right, there are links for "Daftar Produk" and "Log Out". Below the header, there is a section titled "Daftar Produk" with a blue button labeled "Tambah Produk". Below this is a table with the following data:

Nama Produk	Harga	Kategori	Aksi
HP Spectre	Rp 2.500.000	Tanpa Kategori	Edit Hapus
Acer Nitro	Rp 15.000.000	Tanpa Kategori	Edit Hapus
Asus ROG	Rp 15.000.000	Tanpa Kategori	Edit Hapus
Laptop	Rp 20.000.000	Tanpa Kategori	Edit Hapus



c. Penjelasan dari code dan output (korelasi)

Sistem dikembangkan lebih lanjut untuk menangani hubungan relasional *One-to-Many* antara tabel *categories* (Kategori) dan tabel *products* (Produk) melalui modifikasi file migrasi yang menyuntikkan kolom kunci tamu (*foreign key*) berupa *category_id*. Hubungan ini dipetakan di sisi backend melalui fungsi *category()* di dalam Model, dan dipanggil di dalam *controller* menggunakan teknik *eager loading* lewat perintah `Product::with('category')->get()` guna mengoptimalkan kinerja kueri database dan mencegah redundansi data. Hasil dari korelasi relasional ini dapat langsung terlihat secara visual pada halaman output tabel utama di browser, di mana sintaks objek berantai `{{ $product->category->category_name ?? 'Tanpa Kategori' }}` secara otomatis melacak dan menampilkan nama kategori yang cocok bagi setiap produk di dalam kolom baru, serta menampilkan teks default jika produk tersebut belum memiliki kategori.